

Search Relevance&LLM Judge

Melih Durhasan

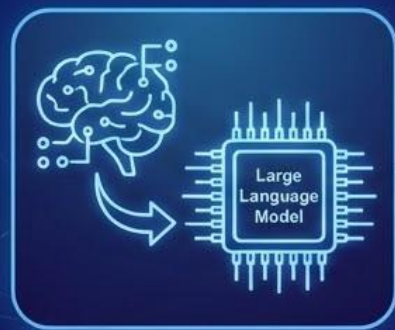
Problem and Goal

- Search engines generate a large number of query–product matches, and it is essential to determine which ones are truly relevant.
- Manual labeling is accurate but slow and costly.
- The solution is to use both traditional search metrics (nDCG, Top-K, MRR) and large language models (LLMs) for automatic labeling and evaluation.



Presentation Flow

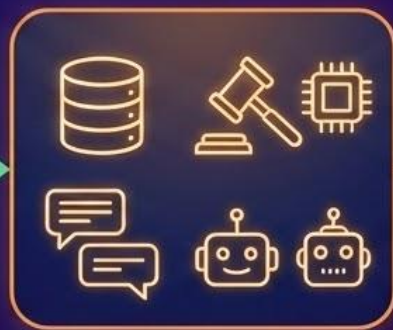
- 1) What is an LLM?
- 2) Mathematical Metrics
- 3) Project Details (ESCI Data, LLM Judge, prompt versions, model comparisons)
- 4) Demo



1. What is an LLM?



2. Mathematical Metrics



3. Project Details
(ESCI Data, LLM Judge, prompt versions, model comparisons)



4. Demo

What is LLM Judge?

Large Language Model (LLM)



An artificial intelligence model trained on very large amounts of text data.



It can perform tasks such as classification, summarization, and reasoning in natural language, and can infer meaning from examples and context.

Role in the Project



Providing the model with a search query and product information



Having the model generate an ESCI label (Exact, Substitute, Complement, Irrelevant) as output



Comparing the results with manual labels and evaluating performance

Veri kaynağı

Veri seç

Amazon ESCI — 1000q



> Amazon ESCI indir

☐ Yalnız Task 1 small_version

CSV/Parquet yükle

Drag and drop file here

Limit 200MB per file • CSV, PARQUET

Browse files

LLM ayarları

LLM Judge — Tek sorgu test

Sorgu

iphone 15 pro max 256 gb siyah

Özellik

Başlık + açıklama



Ürün başlığı

Apple iPhone 15 Pro Max 256GB Siyah Titanyum — Garan

Shot

Zero-shot



Prompt

V1 (baseline)



Ürün açıklaması

6.7 inç Super Retina ekran, A17 Pro çip, 48 MP ana kamera. Kutulu, distribütör garantili.

LLM ile değerlendir

Exact (güven: 1.00)

Veri kaynağı

Veri seç

Amazon ESCI — 1000q

> Amazon ESCI indir

☐ Yalnız Task 1 small_version

CSV/Parquet yükle

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Limit 200MB per file • CSV, PARQUET

Browse files

LLM ayarları

LLM Judge — Tek sorgu test

Sorgu

ipad pro 11 inç tablet

Özellik

Başlık + açıklama

Ürün başlığı

Samsung Galaxy Tab S9 11" 128GB Wi-Fi — Grafite

Shot

Zero-shot

Prompt

V4 (en iyi)

Ürün açıklaması

11 inç Dynamic AMOLED 2X ekran, Snapdragon 8 Gen 2, S Pen kutuda. Android tablet.

LLM ile değerlendir

Substitute (güven: 0.90)

Samsung Galaxy Tab S9 11" 128 GB Wi-Fi_Grafit

Zero-shot

Prompt

V1 (baseline)

Ürün açıklaması

11 inc Dynamic AMOLED 2X ekran, Snapdragon 8 Gen 2, S Pen kutuda. Android tablet.

LLM ile değerlendir

Substitute (güven: 0.80)

Reasoning (CoT)

The query is for an iPad Pro 11 inch tablet, while the product is a Samsung Galaxy Tab S9, which is a different brand and model. Both are tablets with similar sizes, but they are not the same product. Therefore, it is a substitute.

Why is it reliable?

- ❖ Responses are generated in a **structured format** (e.g., JSON) including the **label and a short rationale** (summary of chain-of-thought).
- ❖ The temperature is kept **low** to ensure **consistent** and **reproducible** results.
- ❖ Finally, predictions are compared with **ground truth labels** and evaluated using metrics such as accuracy, F1 score, and confusion matrix.



Structured Output

Responses are generated in a **structured format** (e.g., JSON) including the **label and a short rationale** (summary of chain-of-thought).



Low Temperature

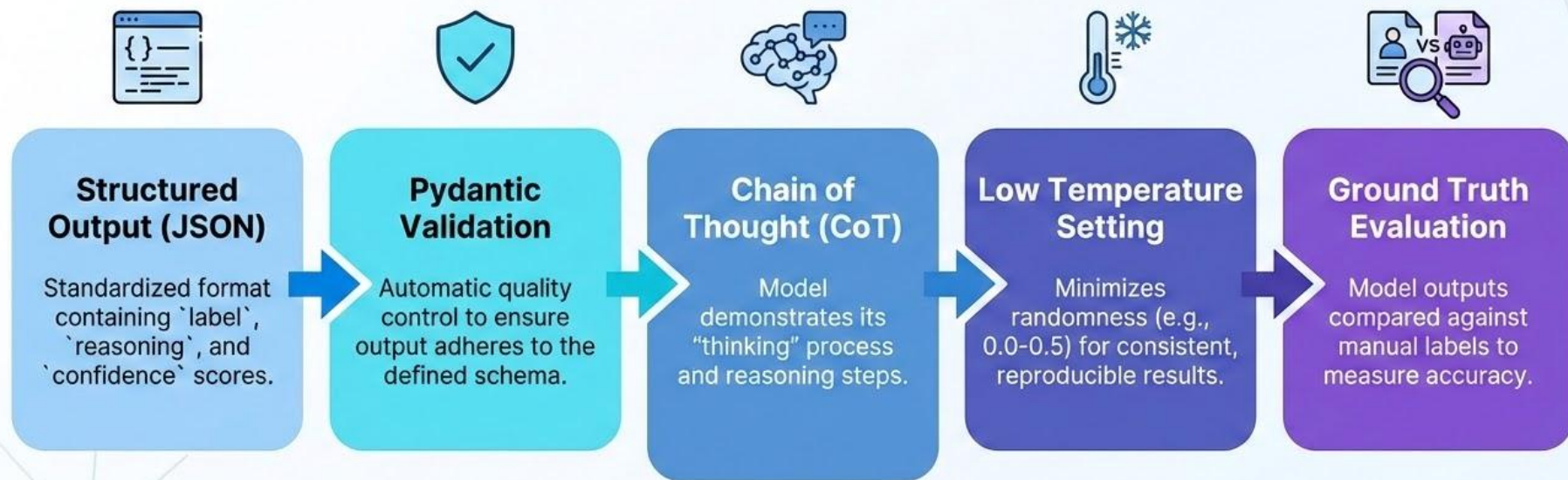
The temperature is kept **low** to ensure **consistent** and **reproducible** results.



Ground Truth Evaluation

Finally, predictions are compared with **ground truth labels** and evaluated using metrics such as accuracy, F1 score, and confusion matrix.

LLM Evaluation Methodology: A Robust Pipeline



Search Metrics

nDCG@K: Measures how well the results are ranked within the top K positions.



Top-K Exact: Checks whether there is at least one *Exact* label within the top K results.



MRR (Mean Reciprocal Rank): Indicates the rank of the first correct result.



Metrics used for LLM evaluation:

- **Accuracy**
- **Precision / Recall / F1 (Macro):** Overall performance across all four classes
- **Class-based F1:** Performance evaluated separately for each class
- **Confusion Matrix**
- **Mean Query Accuracy**

MRR

$$\text{MRR} = \frac{1}{U} \sum_{u=1}^U \frac{1}{\text{rank}_i}$$

NDCG

$$\text{DCG@K} = \sum_{k=1}^K \frac{\text{rel}_i}{\log_2(i+1)}$$

$$\text{NDCG@K} = \frac{\text{DCG@K}}{\text{IDCG@K}}$$



Project Description

Data: Amazon ESCI dataset (query-product pairs with E/S/C/I labels)

Preprocessing: Merging, sampling, optional multi-label query selection

Search Metrics: nDCG, MRR, Top-K (based on ground truth ranking)

LLM Judge: Query + product text → predicted label

Comparison: Prompt versions (V1–V4), model sizes (nano/mini/gpt), shot & feature experiments

Presentation: Streamlit dashboard, packaged with Docker

Data & Preprocessing Workflow



Preprocessing: Merging, sampling, optional multi-label query selection



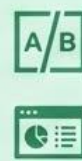
Core System: Search & LLM Judge



Search Metrics: nDCG, text → predicted label
LLM Judge: Query + product



Analysis & Delivery



Comparison: Prompt versions (V1–V4), model sizes (nano/mini/gpt), shot Presentation 

Amazon Data Metrics Results

Veri kaynağı

Veri seç

Amazon ESCI — 1000q

> Amazon ESCI indir

☐ Yalnız Task 1 small_version

CSV/Parquet yükle

Drag and drop file here

Limit 200MB per file • CSV, PARQUET

Browse files

TOP-K & nDCG LLM Judge & Accuracy Prompt & Model Kıyası Error Analysis DataFrame

TOP-K Exact rate & nDCG@k

Exact in TOP-K

	k	Exact rate
0	TOP-1	0.672
1	TOP-3	0.84
2	TOP-4	0.879
3	TOP-5	0.898
4	TOP-10	0.962

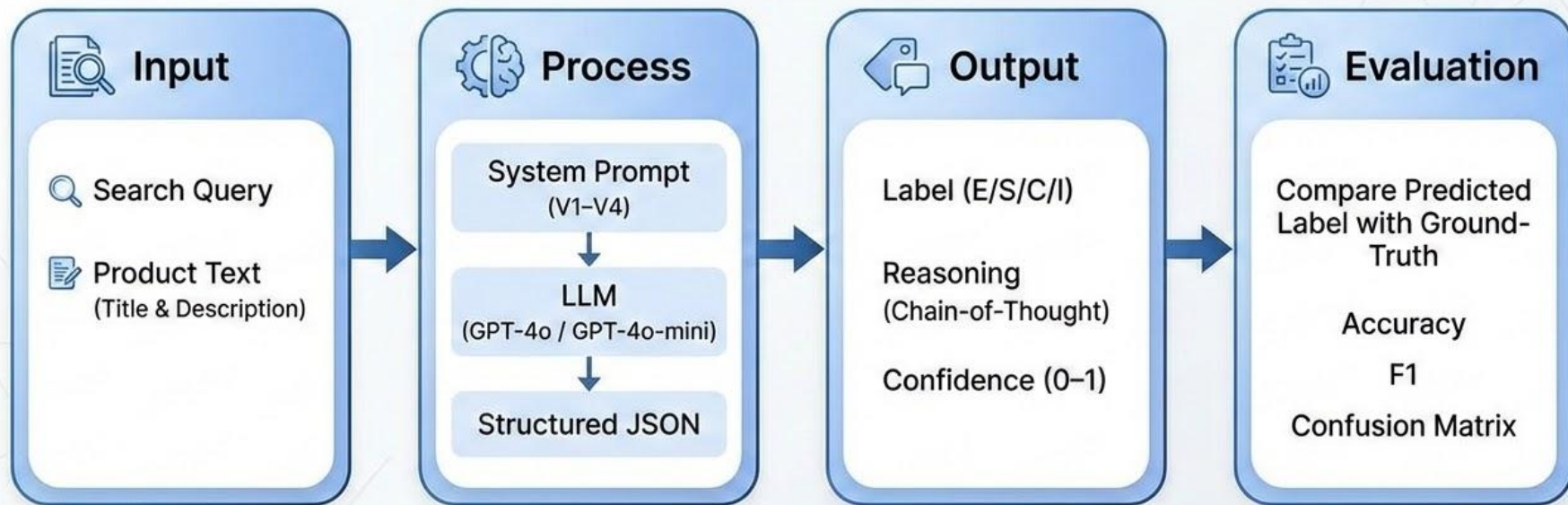
nDCG@k (ortalama)

	k	nDCG
0	nDCG@1	0.899
1	nDCG@3	0.9165
2	nDCG@4	0.9136
3	nDCG@5	0.9107
4	nDCG@10	0.9018

ESCI- Four Relevance Classes

Label	Meaning	Example
Exact	Product directly fulfills the search intent (brand irrelevant).	"running shoes" → Nike Air Max Running Shoe
Substitute	Different product type but satisfies the same underlying need.	"iPad" → Samsung Galaxy Tab
Complement	Accessory or add-on meant to be used together with the item.	"laptop" → Laptop Sleeve
Irrelevant	No logical connection between the query and the product.	"headphones" → Kitchen Towels

LLM Judge - How it Works?



V1 Baseline Sample Search Results:

Toplu LLM vs Ground Truth

Sorgu modu: Aynı sorguda tek ESCI etiketi olanları atlayıp yalnızca karışık sorgularla ölçüm (supervisor önerisi). Satır modu: Verinin en üstten N satırı.

Ölçüm

☐ Satır (üstten N satır) ☒ Sorgu (N sorgu — birden fazla etiketli sorgular)

Sorgu sayısı (çok etiketli)



5

- +

☒ Tek etiketli sorguları çıkar

Accuracy hesapla

Accuracy (satır)	Precision	Recall	F1-macro
48.00%	0.673	0.500	0.453
Ort. sorgu doğruluğu	Sorgu sayısı	Accuracy (Exact hariç)	F1 ort. (S,C,I)
48.00%	5	47.06%	0.382

Confusion Matrix ↗

Satır sayısı N=25. Ground truth dağılımı: Exact=8, Complement=6, Substitute=6, Irrelevant=5

Sayım — satır = gerçek, sütun = tahmin

	Exact	Substitute	Complement	Irrelevant
Exact	4	4	0	0
Substitute	0	2	0	4
Complement	0	0	1	5
Irrelevant	0	0	0	5

Structured Output & Chain-of-Thought



Responses follow a fixed JSON schema: label, reasoning, confidence



No parsing errors — enforced by Pydantic validation



Temperature = 0 for deterministic, reproducible results

Chain-of-Thought (CoT) Process

(1) Query Intent



(2) Product Type

(3) Match Analysis



(4) Label

Model explains its decision step by step before labeling

LLM Judge — Tek sorgu test

Sorgu

iPad Pro 11 inch tablet

Özellik

Başlık + açıklama

Ürün başlığı

Samsung Galaxy Tab S9 11" 128GB Wi-Fi — Graphite

Shot

Zero-shot

Prompt

V4 (en iyi)

Ürün açıklaması

11 inch Dynamic AMOLED 2X display, Snapdragon 8 Gen 2, S Pen included. Android tablet.

LLM ile değerlendir

Substitute (güven: 0.90)

V4 (en iyi)

Ürün açıklaması

11 inch Dynamic AMOLED 2X display, Snapdragon 8 Gen 2, S Pen included. Android tablet.

LLM ile değerlendir

Substitute (güven: 0.90)

Reasoning (CoT)

Step 1: The product (Samsung Galaxy Tab S9) is a tablet, which has a connection to the query (iPad Pro 11 inch tablet). Step 2: The product is not the same type of item as the user searched for because it is a Samsung tablet instead of an iPad. Step 3: The Samsung Galaxy Tab S9 could replace the iPad Pro 11 inch tablet as it serves a similar core need (tablet functionality). Therefore, it is classified as a Substitute.

V4 Sample Search Chain of thought steps:

Feature Ablation- What the Model Sees

Model Input Comparison Table

Mode	Input to the Model	When Useful
Title Only	Only product title	Fast processing; but keep in mind that titles can sometimes be misleading.
Description Only	Only product description	Provides extra context when the title is too vague or lacks detail.
Title + Description	Title + full description	Provides the richest information ; generally yields the best results.

Error Analysis

Bu sekme **hata avcısı** **değil**: seçtiğin satırda LLM'e **aynı soruyu üç kez** sorar (sadece başlık / sadece açıklama / ikisi birden). Kolay örneklerde üçü de **doğru** çıkar; bu normal. Gerçek hatayı görmek için aşağıdan **yanlış tahmin edilen satırları** seç.

Prompt

V1

Satır listesi

- ☒ Tüm satırlar
- ☐ Yalnızca yanlış tahmin (Tab 2 → Accuracy hesapla sonrası)

Satır seç

Satır 6: ...Logitech G502 X...

Error analysis çalıştır

Query: gaming mouse

Product: Logitech G502 X

Doğru etiket: E | Üstteki tek tahmin (Pred): Substitute

Error analysis çalıştır

Query: gaming mouse

Product: Logitech G502 X

Doğru etiket: E | Üstteki tek tahmin (Pred): Substitute

1/2

Arama Alakalılık Paneli

Bu satırda tek tahmin yanlış — aşağıda hangi modda nasıl sapıyor bak.

▼ X title_only: Substitute

The query 'gaming mouse' is looking for a type of product, while 'Logitech G502 X' is a specific model of a gaming mouse. Although it is a relevant product, it is not the exact item searched for, as there are many other brands and models of gaming mice available.

▼ X description_only: Irrelevant

The product description is empty, providing no information about the product. Therefore, it cannot be determined if it relates to the query 'gaming mouse'.

▼ X title_description: Substitute

The query 'gaming mouse' is looking for a type of product, while 'Logitech G502 X' is a specific model of a gaming mouse. Although it is a relevant product, it is not the exact item searched for since there are many other gaming mice available. Therefore, it is considered a substitute rather than an exact match.

Error Analysis tab runs all three on the same row to diagnose where mistakes come from.

Shot Modes: Zero-shot, One-shot, Few-shot

- **Zero-shot:** No examples in the prompt — model relies only on instructions
- **One-shot:** One labeled example included — helps the model understand the format
- **Few-shot:** Multiple examples — strongest teaching signal, more consistent outputs
- In this project: V1–V2 = zero-shot, V3 = one-shot style, V4 = few-shot



Zero-shot

No examples in the prompt — model relies only on instructions



One-shot

One labeled example included — helps the model understand the format



Few-shot

Multiple examples — strongest teaching signal, more consistent outputs

In this project: V1–V2 = zero-shot, V3 = one-shot style, V4 = few-shot

Prompt Engineering: Prompt Versions, V1 to V4

Version	Strategy	Product Info	Shot Mode	Expected Weakness
V1	Weak definitions (brand-sensitive)	Title only	Zero-shot	Marks different brand as Substitute
V2	Corrected ESCI definitions	Title only	Zero-shot	S/C (Substitute/Complement) confusion possible
V3	Detailed rules + critical distinctions	Title + Description	One-shot	Rare edge cases
V4	Decision tree + common pitfalls	Title + Description	Few-shot	Minimal errors

V1 doğruluk (accuracy)

43.8%

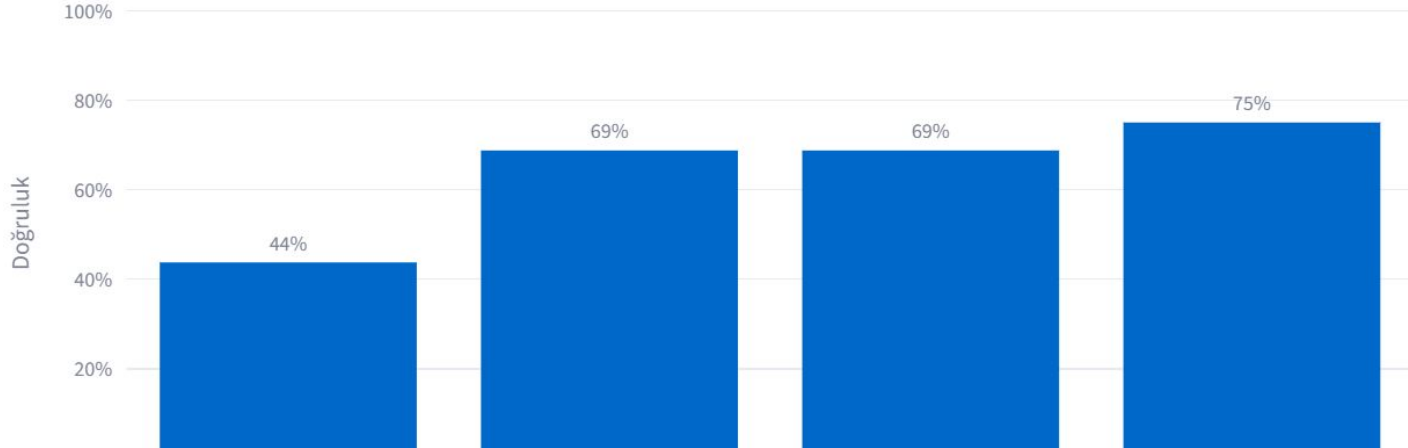
En iyi (V4)

75.0%

↑ +31.2 pp

1) Hangi prompt daha doğru? (accuracy — satır)

Accuracy — yüksek çubuk = ground truth'a daha yakın



Grafikler — 10 sorgu, 50 satır, 4 prompt

V1 doğruluk (accuracy)

42.0%

En iyi (V2)

74.0%

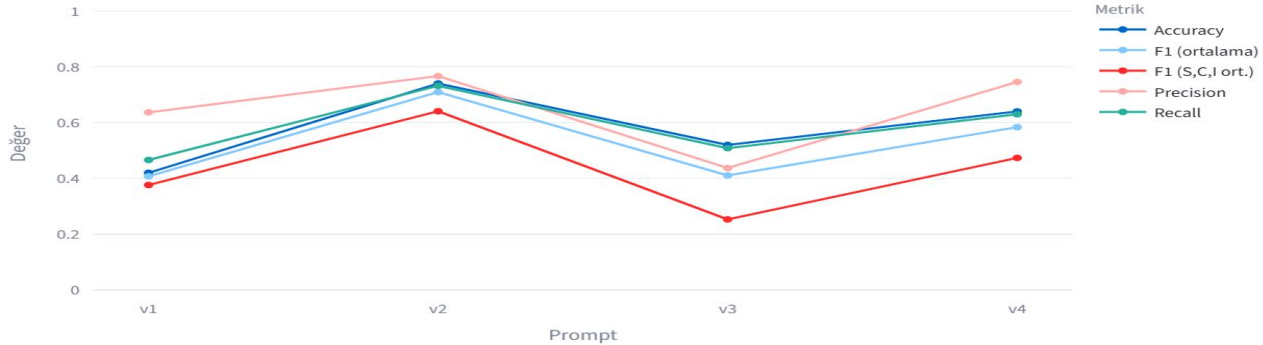
↑ +32.0 pp

1) Hangi prompt daha doğru? (accuracy — satır)

Accuracy — yüksek çubuk = ground truth'a daha yakın

2) V1 → V4 ilerledikçe metrikler

Prompt sürümü arttıkça skorlar



	prompt_version	feature_mode	shot_mode	accuracy	f1_macro	precision_macro	recall_macro	correct_count	total_count
0	v1	title_only	zero_shot	0.42	0.4073	0.6365	0.466	21	50
1	v2	title_only	zero_shot	0.74	0.7091	0.7665	0.7318	37	50
2	v3	title_description	one_shot	0.52	0.4104	0.437	0.5083	26	50
3	v4	title_description	few_shot	0.64	0.5837	0.7457	0.6299	32	50

Model Tiers — Accuracy vs Cost

⋮

Prompt iyileştirme & Model kıyası

Prompt V1→V4 karşılaştırma **Model: nano / mini / gpt + maliyet**

nano / mini / gpt: Aynı satırlar, aynı prompt ile farklı modellerin accuracy ve tahmini maliyetini karşılaştırın. **En verimli** (accuracy / maliyet) ve **en doğru** modeli görün.

> Model eşleşmesi

Prompt

V1

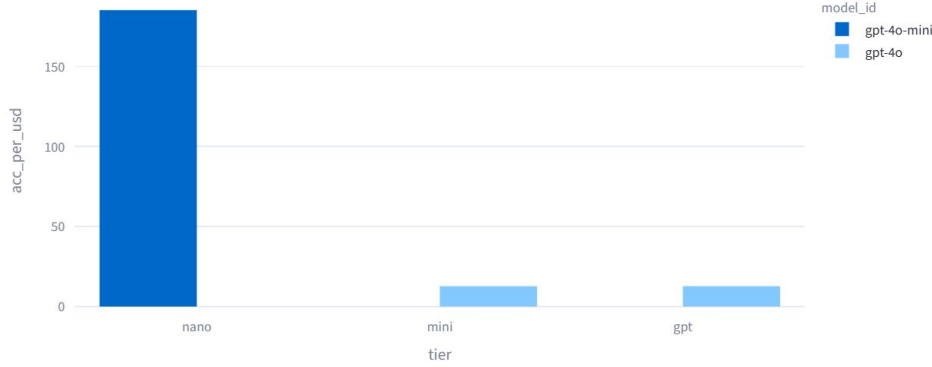
Satır sayısı

10

10 satır × 3 model = 30 çağrı

Nano / mini / gpt çalıştır

Verimlilik: Accuracy / USD (yüksek = daha verimli)



10 lines x3 versions:

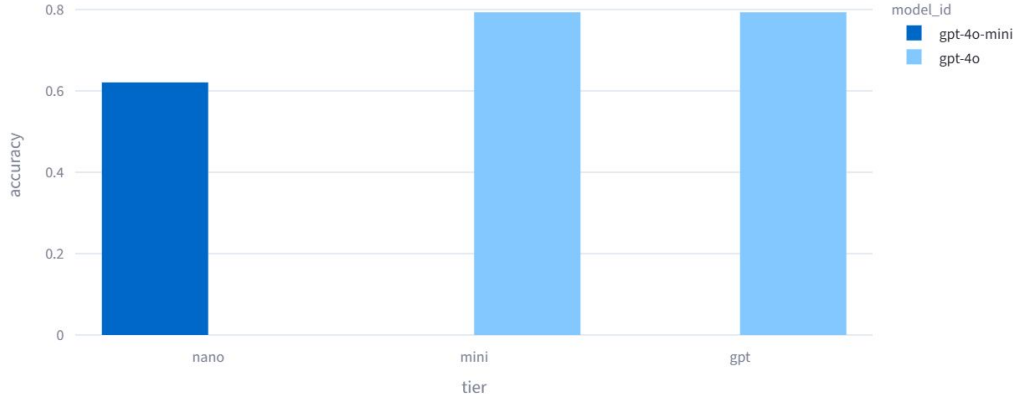
Maliyet tablosu (ChatGPT fiyatları)

	Katman	Model	Input \$/1M	Output \$/1M	Maliyet (\$)	Accuracy %	Doğru/Toplam
0	NANO	gpt-4o-mini	0.15	0.6	0.0038	70	7/10
1	MINI	gpt-4o	2.5	10	0.063	80	8/10
2	GPT	gpt-4o	2.5	10	0.063	80	8/10

En verimli: NANO — 70.0%, \$0.0038

En doğru: MINI — 80.0%, \$0.0630

Accuracy: nano vs mini vs gpt



30 lines x 3 versions.

Maliyet tablosu (ChatGPT fiyatları)

	Katman	Model	Input \$/1M	Output \$/1M	Maliyet (\$)	Accuracy %	Doğru/Toplam
0	NANO	gpt-4o-mini	0.15	0.6	0.011	62.07	18/29
1	MINI	gpt-4o	2.5	10	0.1827	79.31	23/29
2	GPT	gpt-4o	2.5	10	0.1827	79.31	23/29

En verimli: NANO — 62.1%, \$0.0110

En doğru: MINI — 79.3%, \$0.1827

DOCKERIZED DEPLOYMENT



WHAT IS DOCKER?

What is Docker?

- Packages the entire project (code + dependencies) into a container — runs the same way everywhere.

Dockerfile (how the container is built):

- Starts from Python 3.12
- Installs all dependencies (`requirements.txt`)
- Runs Streamlit on port 8501

`docker-compose.yml` (how to launch):

- One command: `docker-compose up --build`
- App ready at `http://localhost:8501`



DOCKER-COMPOSE.YML (how to launch):

- One command: `docker-compose up --build`
- App ready at `http://localhost:8501`

```
[*]: FROM python:3.12-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install -r requirements.txt
COPY . .
EXPOSE 8501
CMD ["streamlit", "run", "streamlit_app.py"]
```

```
[*]: services:
  relevance-ui:
    build: .
    ports:
      - "8501:8501"
    env_file:
      - .env
```

Summary & Key Takeaways

-  Measured search relevance using nDCG, MRR, TOP-K on Amazon ESCI dataset (1000 queries).
-  Built an LLM Judge that automatically labels query–product pairs as Exact / Substitute / Complement / Irrelevant.
-  Used structured output (JSON) and chain-of-thought reasoning for reliable, parseable predictions.
-  Improved accuracy through prompt engineering (V1 → V4): better definitions, more context, few-shot examples.
-  Compared three model tiers (nano / mini / gpt) with cost–efficiency analysis.
-  Experimented with feature modes (title only / description only / both) and shot modes (zero / one / few).
-  Delivered an interactive Streamlit dashboard and Dockerized deployment